

Reliability-Centered Self-Supervised Learning for PCOS Ultrasound Classification Under Duplicate-Aware Evaluation

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PCOS Ultrasound SimCLR + BYOL Duplicate-Aware Splits Reliability Audit

Abstract

Public PCOS ultrasound datasets can support rapid AI research, but duplicated images, pHash near-duplicates, label conflicts, and preprocessing artifacts can inflate headline performance. We evaluate PCOS-XAI ultrasound classification with a reliability-centered protocol: exact and perceptual duplicate auditing, cross-label group exclusion, duplicate-aware train/validation/test splits, supervised transfer baselines, SimCLR and BYOL self-supervised encoders, validation-selected operating points, confidence intervals, calibration, corruption robustness, and Grad-CAM sanity checks. The main finding is nuanced: supervised ResNet-18 remains the strongest low-label default, while SimCLR and BYOL become competitive at 50% labels when threshold selection is handled carefully.

Key Message

Clean accuracy and AUROC are not enough for duplicate-heavy medical imaging datasets.

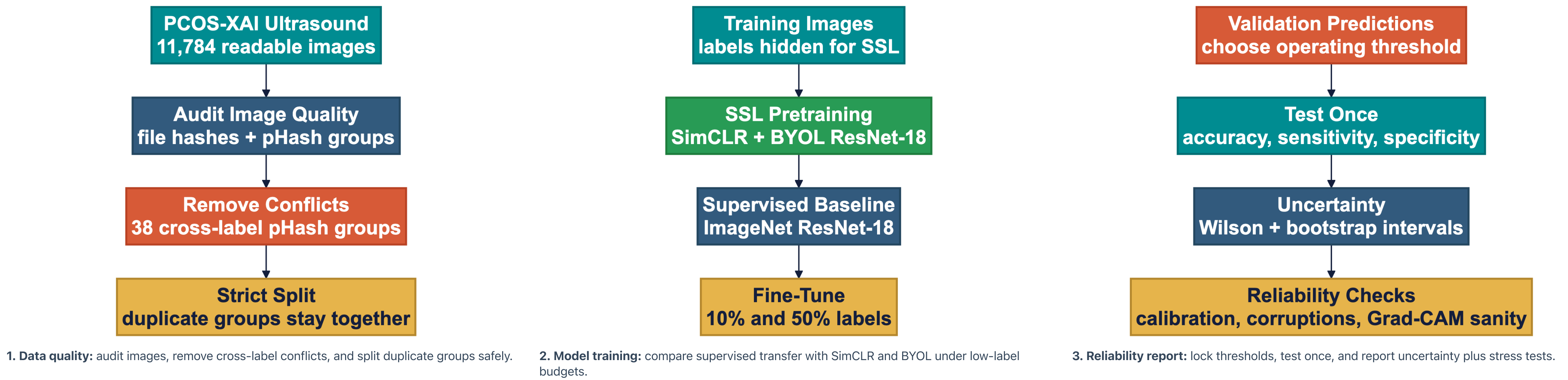
11,784
READABLE IMAGES

0.9939
SIMCLR TUNED ACC.

Duplicate caveat

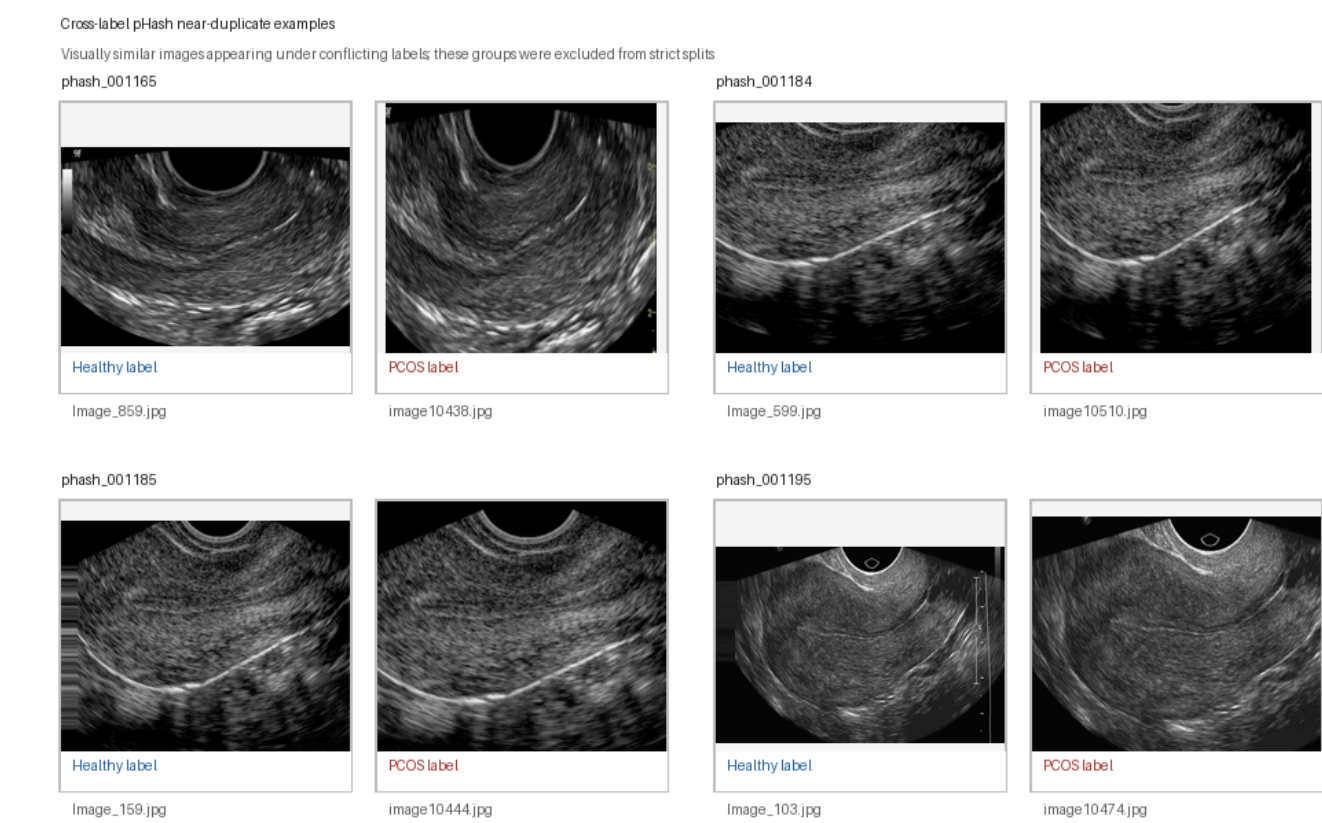
38 cross-label pHash groups were excluded before strict evaluation.

Methodology in 3 Parts



Dataset Audit

6,784 PCOS-positive	5,000 healthy/non-PCOS
1,956 exact duplicate groups	1,788 pHash groups

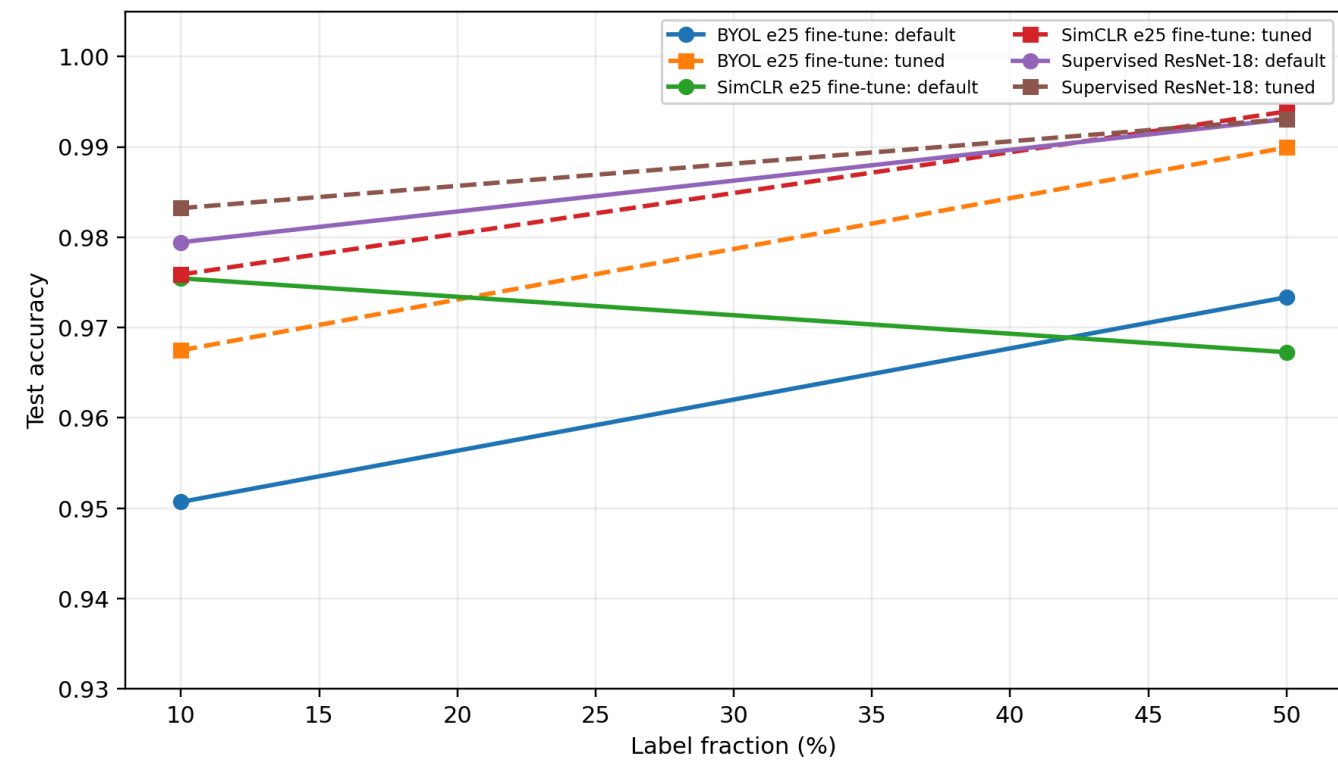


Visually similar images appearing under conflicting labels.

Matched SSL Comparison

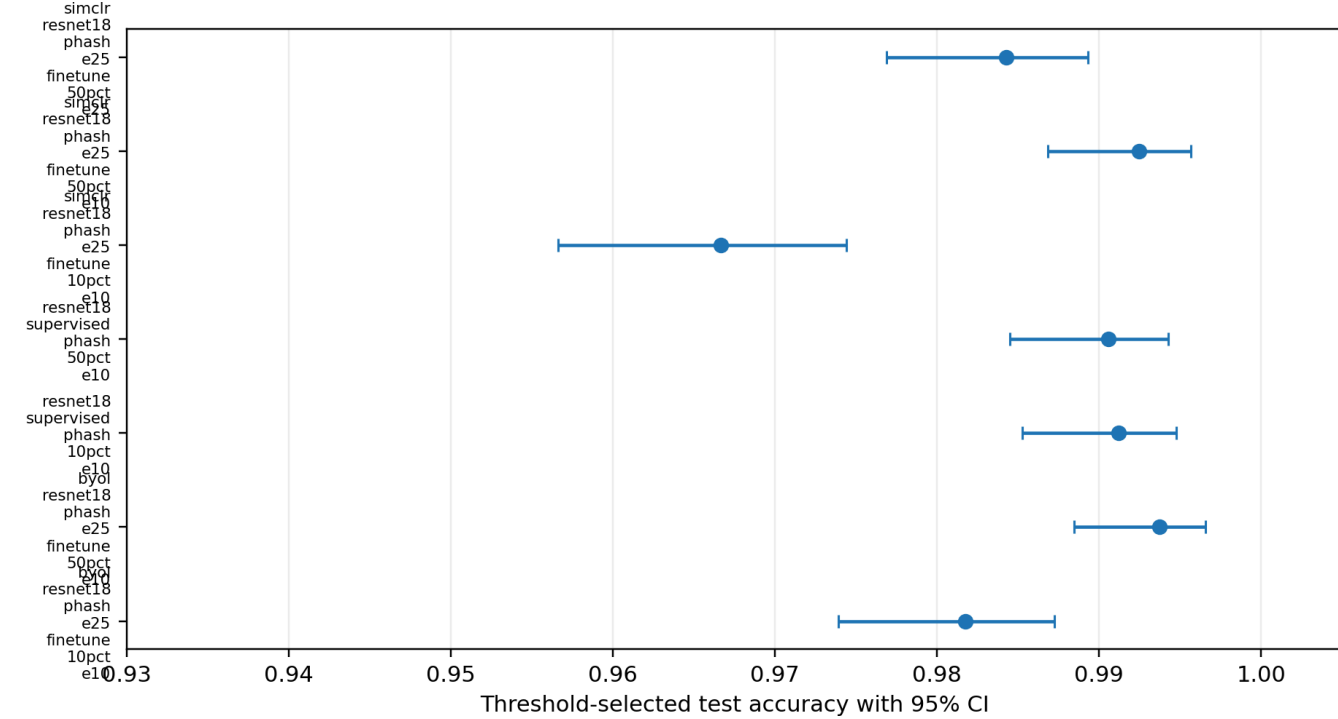
Method	Labels	Acc.	AUROC	Tuned
Supervised ResNet-18	10%	0.9794	0.9984	0.9832
SimCLR e25	10%	0.9755	0.9963	0.9759
BYOL e25	10%	0.9507	0.9942	0.9675
Supervised ResNet-18	50%	0.9931	0.9991	0.9931
SimCLR e25	50%	0.9673	0.9986	0.9939
BYOL e25	50%	0.9734	0.9981	0.9899

Label Efficiency



Operating Points

Checkpoint	Acc.	Sens.	Spec.
Supervised e5	0.9981	0.9966	1.0000
SimCLR e10	0.9924	0.9955	0.9885
BYOL e10	0.9937	0.9955	0.9914



Why Thresholds Matter

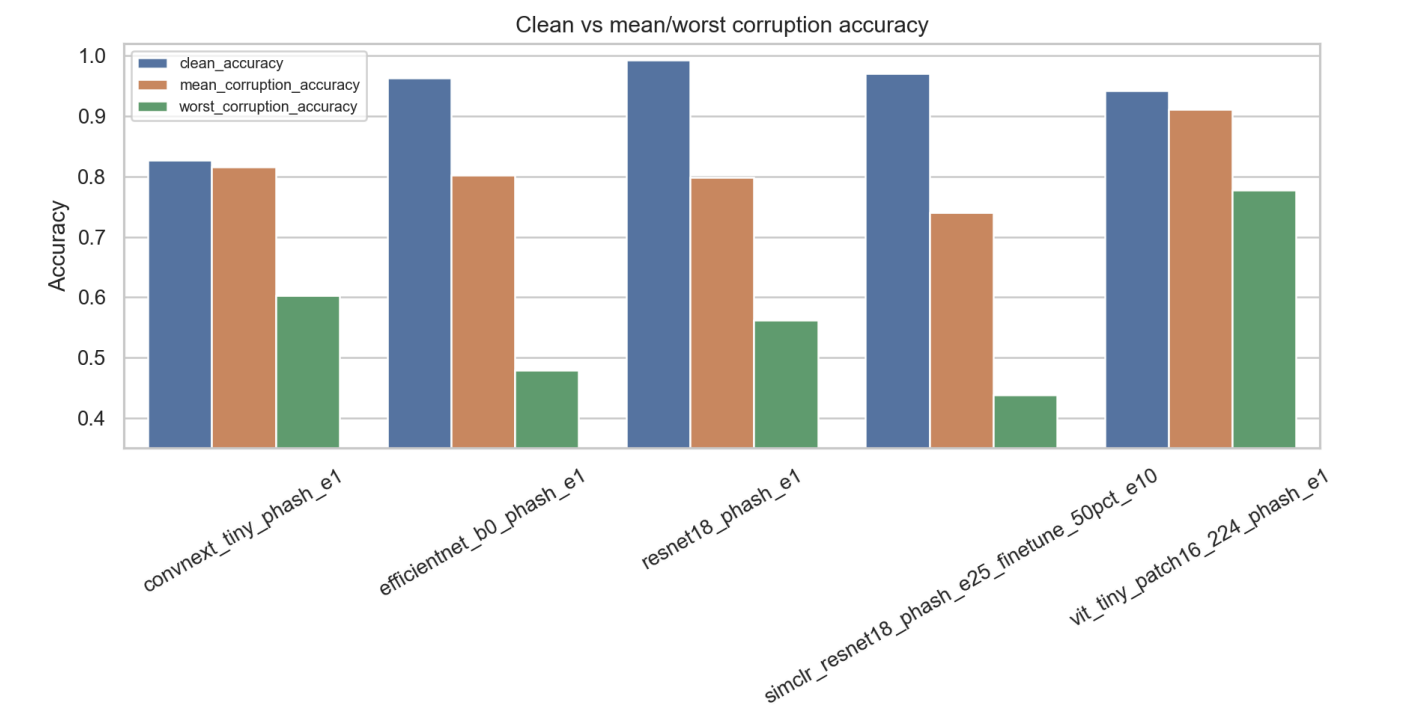
AUROC was almost saturated for strong checkpoints. Validation-selected thresholds better expose sensitivity, specificity, and screening tradeoffs than a default 0.5 threshold alone.

0.9995
supervised AUROC

0.9991
SimCLR AUROC

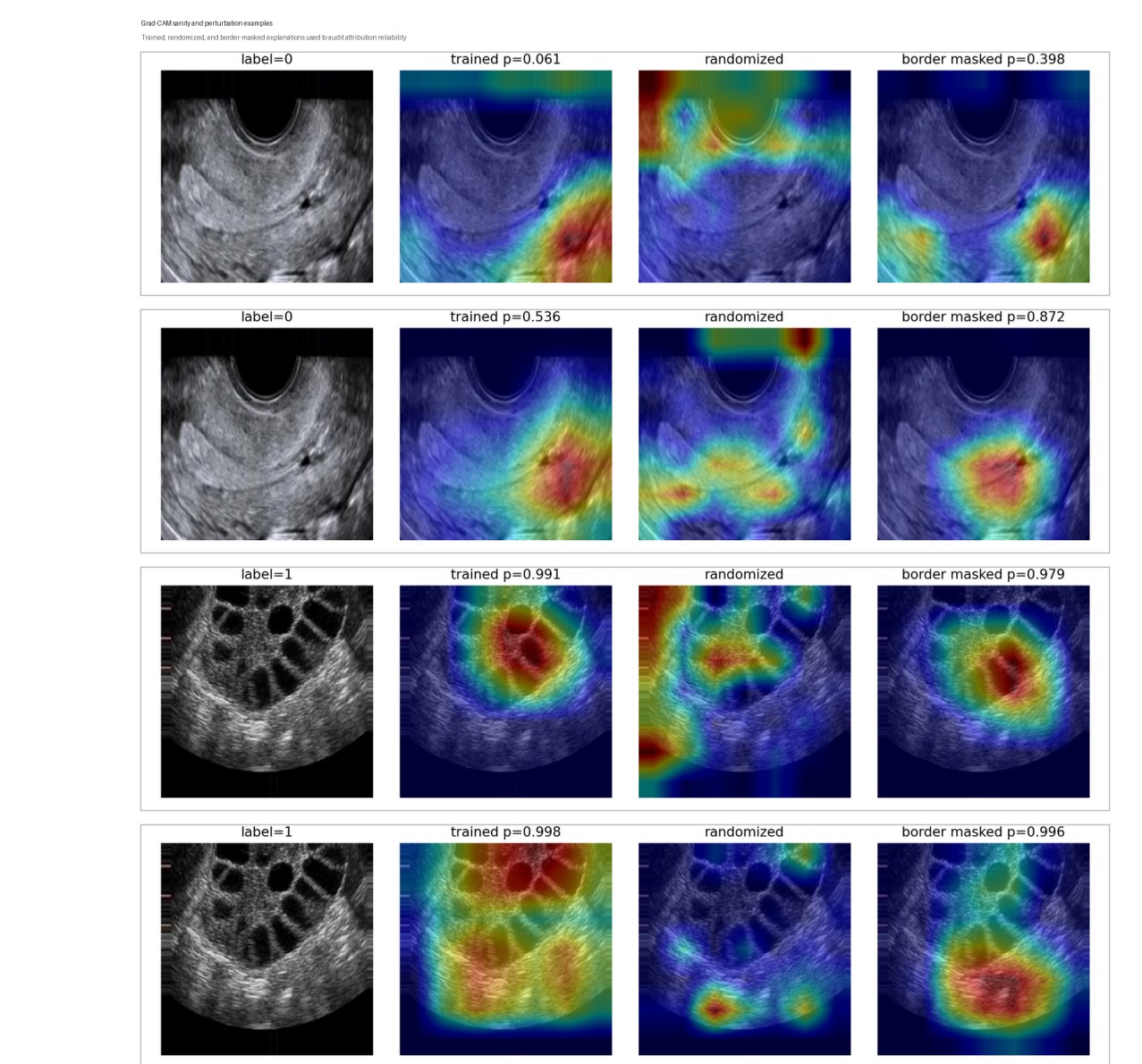
Calibration + Robustness

Calibration improved some checkpoints, but corruption severity showed that clean accuracy did not guarantee stability under blur, downsampling, contrast changes, crop, or noise.



High clean accuracy can still degrade sharply under image corruption.

Grad-CAM Sanity



Interpretation

- Supervised ResNet-18 is the safest low-label default at 10% labels.
- SimCLR and BYOL become competitive at 50% labels after validation thresholding.
- Reliability checks reveal failure modes that clean accuracy hides.
- Grad-CAM should be reported with sanity tests, not as standalone clinical proof.

Conclusion

In low-cleanliness PCOS ultrasound classification, the evaluation protocol is as important as the model: duplicate-aware splitting, confidence intervals, calibration, robustness, and XAI sanity checks make the result more trustworthy than leaderboard-style accuracy alone.